



MOTOROLA
Semiconductors

BOX 20912 • PHOENIX, ARIZONA 85036

MC6820

(0 to 70°C; L or P Suffix)

MC6820C

(-40 to 85°C; L Suffix only)

PERIPHERAL INTERFACE ADAPTER (PIA)

The MC6820 Peripheral Interface Adapter provides the universal means of interfacing peripheral equipment to the MC6800 Microprocessing Unit (MPU). This device is capable of interfacing the MPU to peripherals through two 8-bit bidirectional peripheral data buses and four control lines. No external logic is required for interfacing to most peripheral devices.

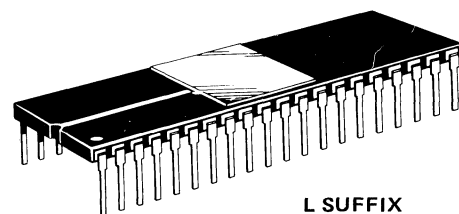
The functional configuration of the PIA is programmed by the MPU during system initialization. Each of the peripheral data lines can be programmed to act as an input or output, and each of the four control/interrupt lines may be programmed for one of several control modes. This allows a high degree of flexibility in the over-all operation of the interface.

- 8-Bit Bidirectional Data Bus for Communication with the MPU
- Two Bidirectional 8-Bit Buses for Interface to Peripherals
- Two Programmable Control Registers
- Two Programmable Data Direction Registers
- Four Individually-Controlled Interrupt Input Lines; Two Usable as Peripheral Control Outputs
- Handshake Control Logic for Input and Output Peripheral Operation
- High-Impedance 3-State and Direct Transistor Drive Peripheral Lines
- Program Controlled Interrupt and Interrupt Disable Capability
- CMOS Drive Capability on Side A Peripheral Lines

MOS

(N-CHANNEL, SILICON-GATE)

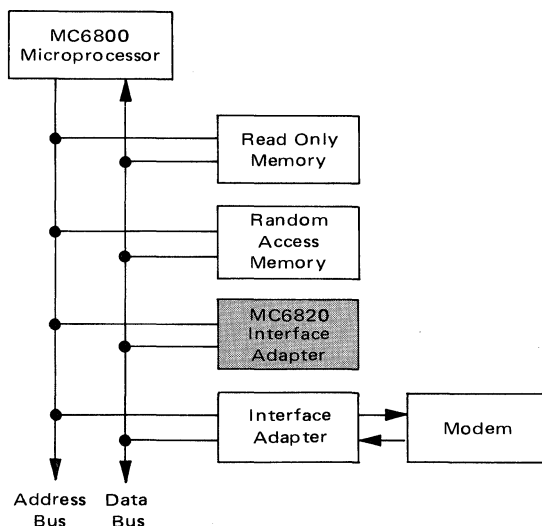
PERIPHERAL INTERFACE ADAPTER



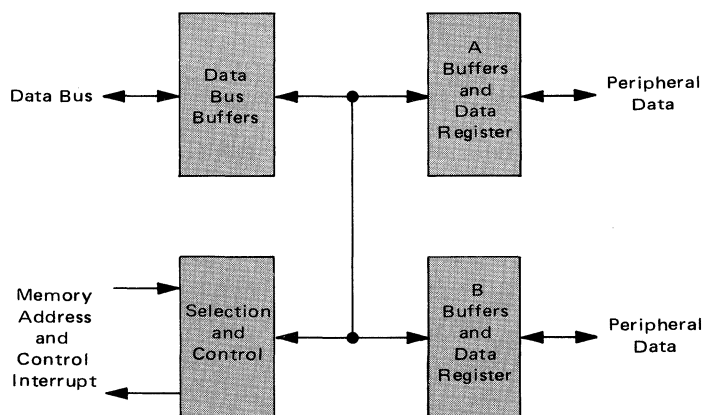
L SUFFIX
CERAMIC PACKAGE
CASE 715

NOT SHOWN: **P SUFFIX**
PLASTIC PACKAGE
CASE 711

**M6800 MICROCOMPUTER FAMILY
BLOCK DIAGRAM**



**MC6820 PERIPHERAL INTERFACE ADAPTER
BLOCK DIAGRAM**



ELECTRICAL CHARACTERISTICS ($V_{CC} = 5.0 \text{ V} \pm 5\%$, $V_{SS} = 0$, $T_A = 0$ to 70°C unless otherwise noted.)

Characteristic	Symbol	Min	Typ	Max	Unit
Input High Voltage Enable Other Inputs	V_{IH}	$V_{SS} + 2.4$ $V_{SS} + 2.0$	— —	V_{CC} V_{CC}	Vdc
Input Low Voltage Enable Other Inputs	V_{IL}	$V_{SS} - 0.3$ $V_{SS} - 0.3$	— —	$V_{SS} + 0.4$ $V_{SS} + 0.8$	Vdc
Input Leakage Current ($V_{in} = 0$ to 5.25 Vdc) R/W, $\overline{\text{Reset}}$, RS0, RS1, CS0, CS1, $\overline{\text{CS2}}$, CA1, CB1, Enable	I_{in}	—	1.0	2.5	μAdc
Three-State (Off State) Input Current ($V_{in} = 0.4$ to 2.4 Vdc) D0-D7, PB0-PB7, CB2	I_{TSI}	—	2.0	10	μAdc
Input High Current ($V_{IH} = 2.4 \text{ Vdc}$) PA0-PA7, CA2	I_{IH}	-100	-250	—	μAdc
Input Low Current ($V_{IL} = 0.4 \text{ Vdc}$) PA0-PA7, CA2	I_{IL}	—	-1.0	-1.6	mAdc
Output High Voltage ($I_{Load} = -205 \mu\text{Adc}$, Enable Pulse Width $< 25 \mu\text{s}$) ($I_{Load} = -100 \mu\text{Adc}$, Enable Pulse Width $< 25 \mu\text{s}$) D0-D7 Other Outputs	V_{OH}	$V_{SS} + 2.4$ $V_{SS} + 2.4$	— —	— —	Vdc
Output Low Voltage ($I_{Load} = 1.6 \text{ mAdc}$, Enable Pulse Width $< 25 \mu\text{s}$)	V_{OL}	—	—	$V_{SS} + 0.4$	Vdc
Output High Current (Sourcing) ($V_{OH} = 2.4 \text{ Vdc}$) D0-D7 Other Outputs ($V_O = 1.5 \text{ Vdc}$, the current for driving other than TTL, e.g., Darlington Base) PB0-PB7, CB2	I_{OH}	-205 -100 -1.0	— — -2.5	— — -10	μAdc μAdc mAdc
Output Low Current (Sinking) ($V_{OL} = 0.4 \text{ Vdc}$)	I_{OL}	1.6	—	—	mAdc
Output Leakage Current (Off State) ($V_{OH} = 2.4 \text{ Vdc}$) $\overline{\text{IRQA}}$, $\overline{\text{IRQB}}$	I_{LOH}	—	1.0	10	μAdc
Power Dissipation	P_D	—	—	650	mW
Input Capacitance ($V_{in} = 0$, $T_A = 25^\circ\text{C}$, $f = 1.0 \text{ MHz}$) Enable D0-D7 PA0-PA7, PB0-PB7, CA2, CB2 R/W, $\overline{\text{Reset}}$, RS0, RS1, CS0, CS1, $\overline{\text{CS2}}$, CA1, CB1	C_{in}	— — — —	— — — —	20 12.5 10 7.5	pF
Output Capacitance ($V_{in} = 0$, $T_A = 25^\circ\text{C}$, $f = 1.0 \text{ MHz}$) $\overline{\text{IRQA}}$, $\overline{\text{IRQB}}$ PB0-PB7	C_{out}	— —	— —	5.0 10	pF
Peripheral Data Setup Time (Figure 1)	t_{PDSU}	200	—	—	ns
Delay Time, Enable negative transition to CA2 negative transition (Figure 2, 3)	t_{CA2}	—	—	1.0	μs
Delay Time, Enable negative transition to CA2 positive transition (Figure 2)	t_{RS1}	—	—	1.0	μs
Rise and Fall Times for CA1 and CA2 input signals (Figure 3)	$t_{r,tf}$	—	—	1.0	μs
Delay Time from CA1 active transition to CA2 positive transition (Figure 3)	t_{RS2}	—	—	2.0	μs
Delay Time, Enable negative transition to Peripheral Data valid (Figures 4, 5)	t_{PDW}	—	—	1.0	μs
Delay Time, Enable negative transition to Peripheral CMOS Data Valid ($V_{CC} - 30\% V_{CC}$, Figure 4; Figure 12 Load C) PA0-PA7, CA2	t_{CMOS}	—	—	2.0	μs
Delay Time, Enable positive transition to CB2 negative transition (Figure 6, 7)	t_{CB2}	—	—	1.0	μs
Delay Time, Peripheral Data valid to CB2 negative transition (Figure 5)	t_{DC}	20	—	—	ns
Delay Time, Enable positive transition to CB2 positive transition (Figure 6)	t_{RS1}	—	—	1.0	μs
Rise and Fall Time for CB1 and CB2 input signals (Figure 7)	$t_{r,tf}$	—	—	1.0	μs
Delay Time, CB1 active transition to CB2 positive transition (Figure 7)	t_{RS2}	—	—	2.0	μs
Interrupt Release Time, $\overline{\text{IRQA}}$ and $\overline{\text{IRQB}}$ (Figure 8)	t_{IR}	—	—	1.6	μs
Reset Low Time* (Figure 9)	t_{RL}	2.0	—	—	μs

*The Reset line must be high a minimum of $1.0 \mu\text{s}$ before addressing the PIA.

MAXIMUM RATINGS

Rating	Symbol	Value	Unit
Supply Voltage	V_{CC}	-0.3 to +7.0	Vdc
Input Voltage	V_{in}	-0.3 to +7.0	Vdc
Operating Temperature Range	T_A	0 to +70	°C
Storage Temperature Range	T_{stg}	-55 to +150	°C
Thermal Resistance	θ_{JA}	82.5	°C/W

This device contains circuitry to protect the inputs against damage due to high static voltages or electric fields; however, it is advised that normal precautions be taken to avoid application of any voltage higher than maximum rated voltages to this high impedance circuit.

BUS TIMING CHARACTERISTICS

READ (Figures 10 and 12)

Characteristic	Symbol	Min	Typ	Max	Unit
Enable Cycle Time	t_{cycE}	1.0	—	—	μs
Enable Pulse Width, High	PW_{EH}	0.45	—	25	μs
Enable Pulse Width, Low	PW_{EL}	0.43	—	—	μs
Setup Time, Address and R/W valid to Enable positive transition	t_{AS}	160	—	—	ns
Data Delay Time	t_{DDR}	—	—	320	ns
Data Hold Time	t_H	10	—	—	ns
Address Hold Time	t_{AH}	10	—	—	ns
Rise and Fall Time for Enable input	t_{Er}, t_{Ef}	—	—	25	ns

WRITE (Figures 11 and 12)

Enable Cycle Time	t_{cycE}	1.0	—	—	μs
Enable Pulse Width, High	PW_{EH}	0.45	—	25	μs
Enable Pulse Width, Low	PW_{EL}	0.43	—	—	μs
Setup Time, Address and R/W valid to Enable positive transition	t_{AS}	160	—	—	ns
Data Setup Time	t_{DSW}	195	—	—	ns
Data Hold Time	t_H	10	—	—	ns
Address Hold Time	t_{AH}	10	—	—	ns
Rise and Fall Time for Enable input	t_{Er}, t_{Ef}	—	—	25	ns

FIGURE 1 — PERIPHERAL DATA SETUP TIME
(Read Mode)

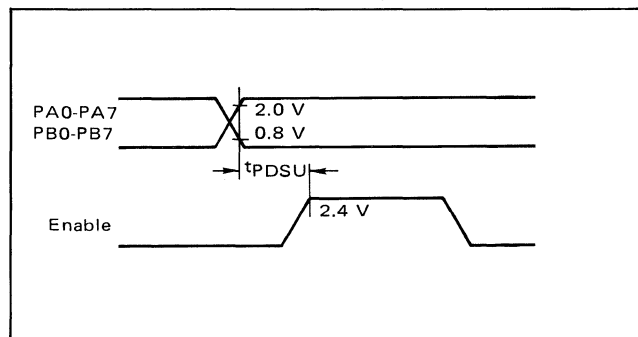


FIGURE 2 — CA2 DELAY TIME
(Read Mode; CRA-5 = CRA-3 = 1, CRA-4 = 0)

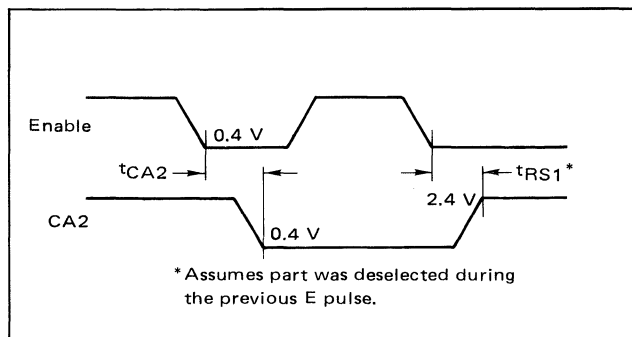


FIGURE 3 — CA2 DELAY TIME
(Read Mode; CRA-5 = 1, CRA-3 = CRA-4 = 0)

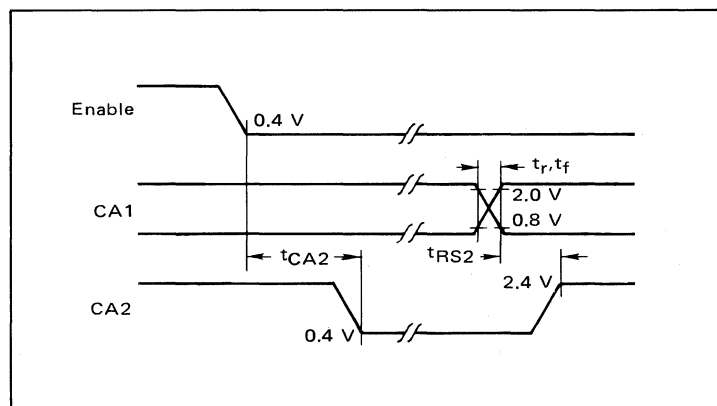


FIGURE 4 – PERIPHERAL CMOS DATA DELAY TIMES
(Write Mode; CRA-5 = CRA-3 = 1, CRA-4 = 0)

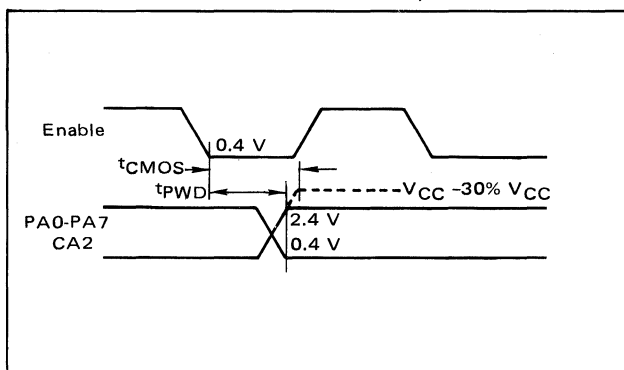


FIGURE 6 – CB2 DELAY TIME
(Write Mode; CRB-5 = CRB-3 = 1, CRB-4 = 0)

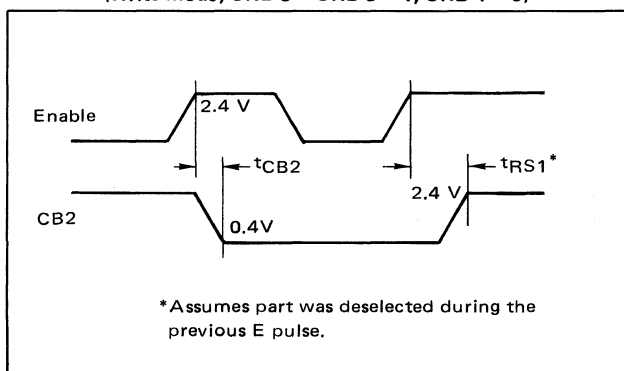


FIGURE 8 – $\overline{\text{TRQ}}$ RELEASE TIME

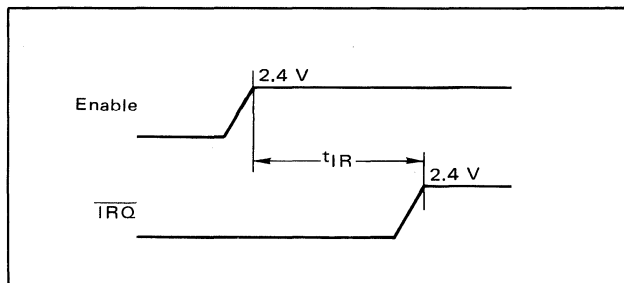


FIGURE 10 – BUS READ TIMING CHARACTERISTICS
(Read Information from PIA)

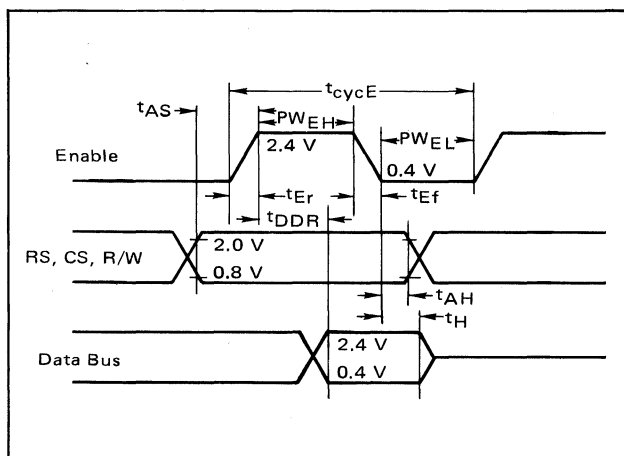


FIGURE 5 – PERIPHERAL DATA AND CB2 DELAY TIMES
(Write Mode; CRB-5 = CRB-3 = 1, CRB-4 = 0)

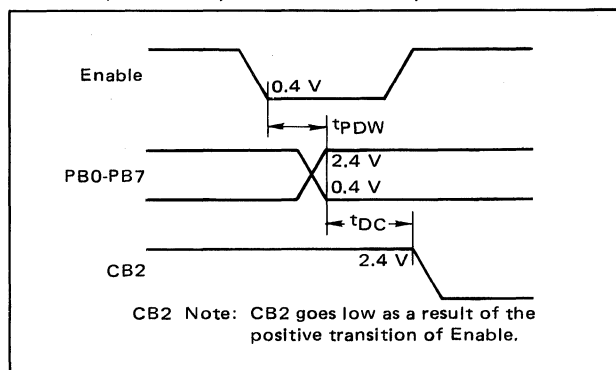


FIGURE 7 – CB2 DELAY TIME
(Write Mode; CRB-5 = 1, CRB-3 = CRB-4 = 0)

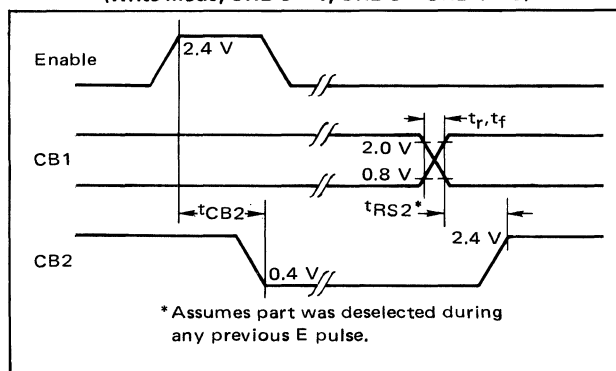


FIGURE 9 – $\overline{\text{RESET}}$ LOW TIME

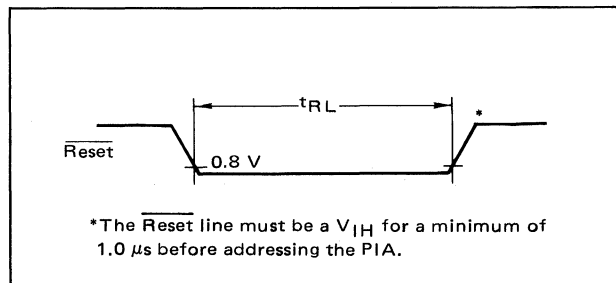


FIGURE 11 – BUS WRITE TIMING CHARACTERISTICS
(Write Information into PIA)

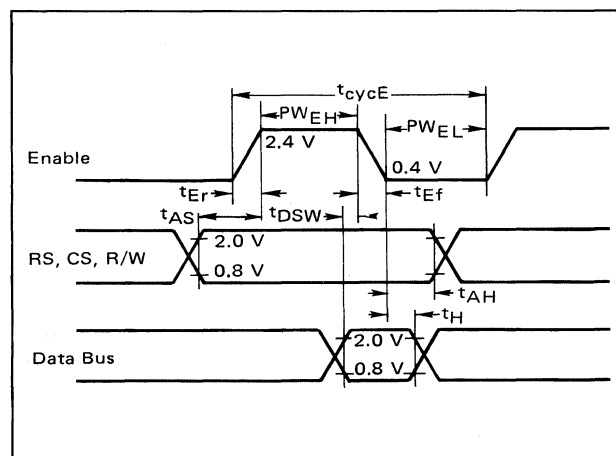
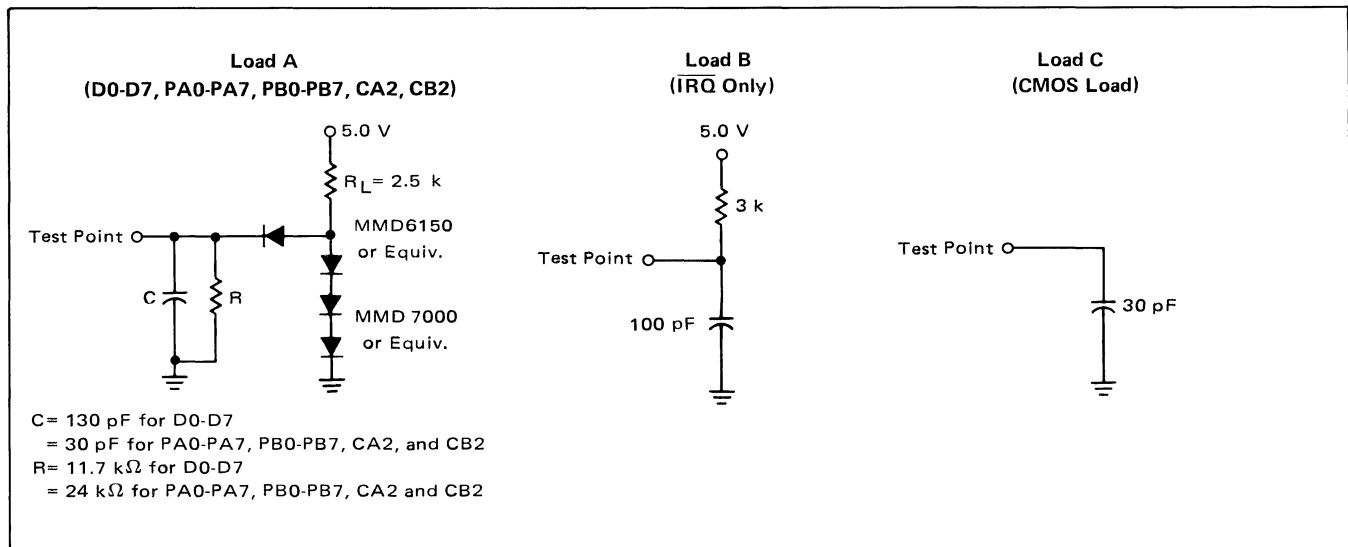


FIGURE 12 – BUS TIMING TEST LOADS



PIA INTERFACE SIGNALS FOR MPU

The PIA interfaces to the MC6800 MPU with an eight-bit bi-directional data bus, three chip select lines, two register select lines, two interrupt request lines, read/write line, enable line and reset line. These signals, in conjunction with the MC6800 VMA output, permit the MPU to have complete control over the PIA. VMA should be utilized in conjunction with an MPU address line into a chip select of the PIA.

PIA Bi-Directional Data (D0-D7) — The bi-directional data lines (D0-D7) allow the transfer of data between the MPU and the PIA. The data bus output drivers are three-state devices that remain in the high-impedance (off) state except when the MPU performs a PIA read operation. The Read/Write line is in the Read (high) state when the PIA is selected for a Read operation.

PIA Enable (E) — The enable pulse, E, is the only timing signal that is supplied to the PIA. Timing of all other signals is referenced to the leading and trailing edges of the E pulse. This signal will normally be a derivative of the MC6800 $\phi 2$ Clock.

PIA Read/Write (R/W) — This signal is generated by the MPU to control the direction of data transfers on the Data Bus. A low state on the PIA Read/Write line enables the input buffers and data is transferred from the MPU to the PIA on the E signal if the device has been selected. A high on the Read/Write line sets up the PIA for a transfer of data to the bus. The PIA output buffers are enabled when the proper address and the enable pulse E are present.

Reset — The active low $\overline{\text{Reset}}$ line is used to reset all register bits in the PIA to a logical zero (low). This line can be used as a power-on reset and as a master reset during system operation.

PIA Chip Select (CS0, CS1 and $\overline{\text{CS2}}$) — These three input signals are used to select the PIA. CS0 and CS1 must be high and $\overline{\text{CS2}}$ must be low for selection of the device. Data transfers are then performed under the control of the Enable and Read/Write signals. The chip select lines must be stable for the duration of the E pulse. The device is deselected when any of the chip selects are in the inactive state.

PIA Register Select (RS0 and RS1) — The two register select lines are used to select the various registers inside the PIA. These two lines are used in conjunction with internal Control Registers to select a particular register that is to be written or read.

The register and chip select lines should be stable for the duration of the E pulse while in the read or write cycle.

Interrupt Request ($\overline{\text{IRQA}}$ and $\overline{\text{IRQB}}$) — The active low Interrupt Request lines ($\overline{\text{IRQA}}$ and $\overline{\text{IRQB}}$) act to interrupt the MPU either directly or through interrupt priority circuitry. These lines are "open drain" (no load device on the chip). This permits all interrupt request lines to be tied together in a wire-OR configuration.

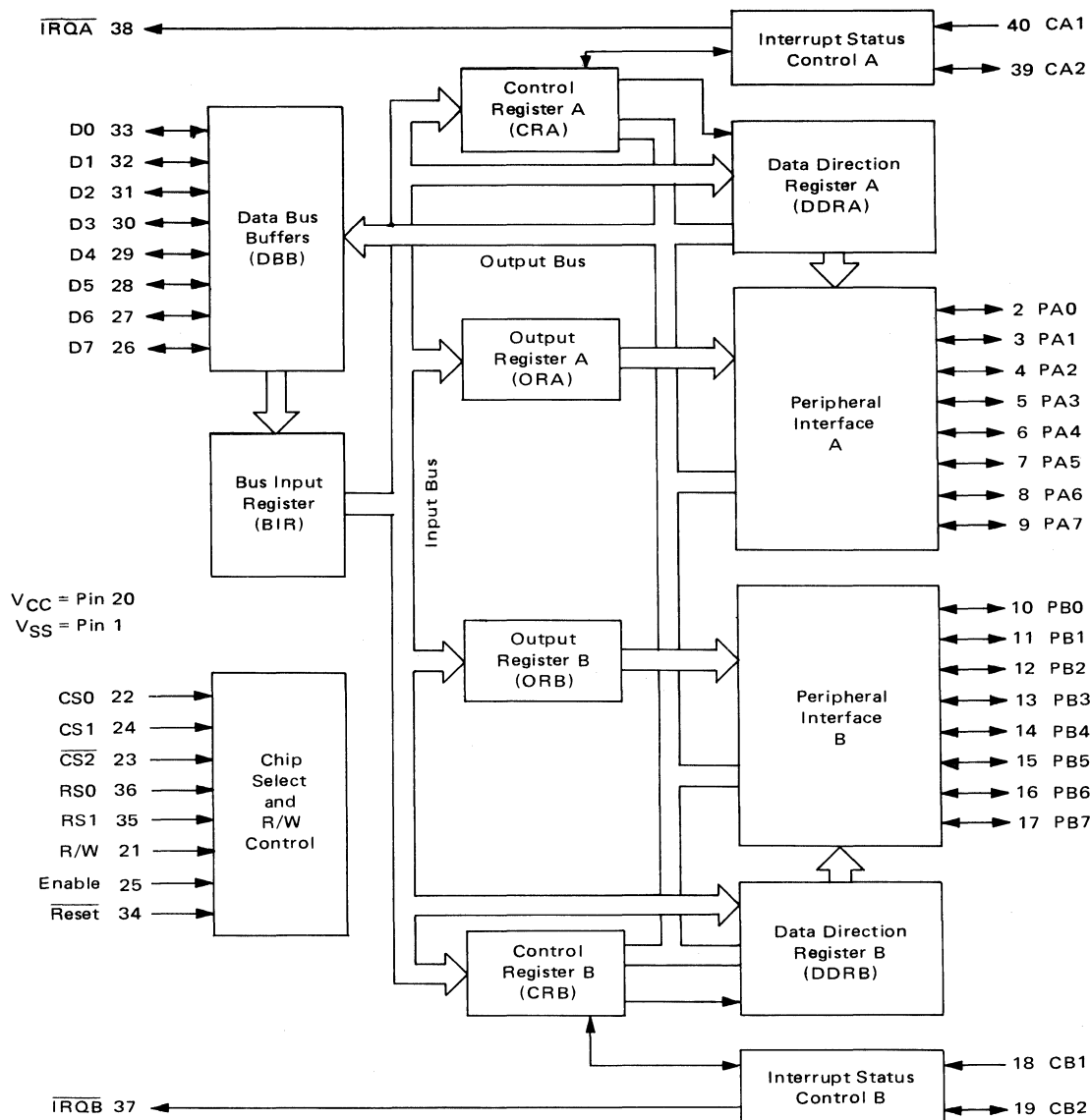
Each Interrupt Request line has two internal interrupt flag bits that can cause the Interrupt Request line to go low. Each flag bit is associated with a particular peripheral interrupt line. Also four interrupt enable bits are provided in the PIA which may be used to inhibit a particular interrupt from a peripheral device.

Servicing an interrupt by the MPU may be accomplished by a software routine that, on a prioritized basis, sequentially reads and tests the two control registers in each PIA for interrupt flag bits that are set.

The interrupt flags are cleared (zeroed) as a result of an



EXPANDED BLOCK DIAGRAM



MPU Read Peripheral Data Operation of the corresponding data register. After being cleared, the interrupt flag bit cannot be enabled to be set until the PIA is deselected during an E pulse. The E pulse is used to condition the interrupt control lines (CA1, CA2, CB1, CB2). When these lines are used as interrupt inputs at least one E

pulse must occur from the inactive edge to the active edge of the interrupt input signal to condition the edge sense network. If the interrupt flag has been enabled and the edge sense circuit has been properly conditioned, the interrupt flag will be set on the next active transition of the interrupt input pin.

PIA PERIPHERAL INTERFACE LINES

The PIA provides two 8-bit bi-directional data buses and four interrupt/control lines for interfacing to peripheral devices.

Section A Peripheral Data (PA0-PA7) — Each of the peripheral data lines can be programmed to act as an input or output. This is accomplished by setting a "1" in the corresponding Data Direction Register bit for those lines which are to be outputs. A "0" in a bit of the Data Direction Register causes the corresponding peripheral data line to act as an input. During an MPU Read Peripheral Data Operation, the data on peripheral lines programmed to act as inputs appears directly on the corresponding MPU Data Bus lines. In the input mode the internal pullup resistor on these lines represents a maximum of one standard TTL load.

The data in Output Register A will appear on the data lines that are programmed to be outputs. A logical "1" written into the register will cause a "high" on the corresponding data line while a "0" results in a "low". Data in Output Register A may be read by an MPU "Read Peripheral Data A" operation when the corresponding lines are programmed as outputs. This data will be read properly if the voltage on the peripheral data lines is greater than 2.0 volts for a logic "1" output and less than 0.8 volt for a logic "0" output. Loading the output lines such that the voltage on these lines does not reach full voltage causes the data transferred into the MPU on a Read operation to differ from that contained in the respective bit of Output Register A.

Section B Peripheral Data (PB0-PB7) — The peripheral data lines in the B Section of the PIA can be programmed to act as either inputs or outputs in a similar manner to PA0-PA7. However, the output buffers driving these lines differ from those driving lines PA0-PA7. They have three-

state capability, allowing them to enter a high impedance state when the peripheral data line is used as an input. In addition, data on the peripheral data lines PB0-PB7 will be read properly from those lines programmed as outputs even if the voltages are below 2.0 volts for a "high". As outputs, these lines are compatible with standard TTL and may also be used as a source of up to 1 milliampere at 1.5 volts to directly drive the base of a transistor switch.

Interrupt Input (CA1 and CB1) — Peripheral Input lines CA1 and CB1 are input only lines that set the interrupt flags of the control registers. The active transition for these signals is also programmed by the two control registers.

Peripheral Control (CA2) — The peripheral control line CA2 can be programmed to act as an interrupt input or as a peripheral control output. As an output, this line is compatible with standard TTL; as an input the internal pullup resistor on this line represents one standard TTL load. The function of this signal line is programmed with Control Register A.

Peripheral Control (CB2) — Peripheral Control line CB2 may also be programmed to act as an interrupt input or peripheral control output. As an input, this line has high input impedance and is compatible with standard TTL. As an output it is compatible with standard TTL and may also be used as a source of up to 1 milliampere at 1.5 volts to directly drive the base of a transistor switch. This line is programmed by Control Register B.

NOTE: It is recommended that the control lines (CA1, CA2, CB1, CB2) should be held in a logic 1 state when $\overline{\text{Reset}}$ is active to prevent setting of corresponding interrupt flags in the control register when $\overline{\text{Reset}}$ goes to an inactive state. Subsequent to $\overline{\text{Reset}}$ going inactive, a read of the data registers may be used to clear any undesired interrupt flags.



INTERNAL CONTROLS

There are six locations within the PIA accessible to the MPU data bus: two Peripheral Registers, two Data Direction Registers, and two Control Registers. Selection of these locations is controlled by the RS0 and RS1 inputs together with bit 2 in the Control Register, as shown in Table 1.

TABLE 1 — INTERNAL ADDRESSING

RS1	RS0	Control Register Bit		Location Selected
		CRA-2	CRB-2	
0	0	1	X	Peripheral Register A
0	0	0	X	Data Direction Register A
0	1	X	X	Control Register A
1	0	X	1	Peripheral Register B
1	0	X	0	Data Direction Register B
1	1	X	X	Control Register B

X = Don't Care

INITIALIZATION

A low reset line has the effect of zeroing all PIA registers. This will set PA0-PA7, PB0-PB7, CA2 and CB2 as inputs, and all interrupts disabled. The PIA must be configured during the restart program which follows the reset.

Details of possible configurations of the Data Direction and Control Register are as follows.

DATA DIRECTION REGISTERS (DDRA and DDRB)

The two Data Direction Registers allow the MPU to control the direction of data through each corresponding peripheral data line. A Data Direction Register bit set at "0" configures the corresponding peripheral data line as an input; a "1" results in an output.

CONTROL REGISTERS (CRA and CRB)

The two Control Registers (CRA and CRB) allow the MPU to control the operation of the four peripheral control lines CA1, CA2, CB1 and CB2. In addition they allow the MPU to enable the interrupt lines and monitor the status of the interrupt flags. Bits 0 through 5 of the two registers may be written or read by the MPU when the proper chip select and register select signals are applied. Bits 6 and 7 of the two registers are read only and are modified by external interrupts occurring on control lines CA1, CA2, CB1 or CB2. The format of the control words is shown in Table 2.

TABLE 2 — CONTROL WORD FORMAT

	7	6	5	4	3	2	1	0
CRA	IRQA1	IRQA2	CA2 Control			DDRA Access	CA1 Control	
CRB	IRQB1	IRQB2	CB2 Control			DDRB Access	CB1 Control	

Data Direction Access Control Bit (CRA-2 and CRB-2) —

Bit 2 in each Control register (CRA and CRB) allows selection of either a Peripheral Interface Register or the Data Direction Register when the proper register select signals are applied to RS0 and RS1.

Interrupt Flags (CRA-6, CRA-7, CRB-6, and CRB-7) —

The four interrupt flag bits are set by active transitions of signals on the four Interrupt and Peripheral Control lines when those lines are programmed to be inputs. These bits cannot be set directly from the MPU Data Bus and are reset indirectly by a Read Peripheral Data Operation on the appropriate section.

TABLE 3 — CONTROL OF INTERRUPT INPUTS CA1 AND CB1

CRA-1 (CRB-1)	CRA-0 (CRB-0)	Interrupt Input CA1 (CB1)	Interrupt Flag CRA-7 (CRB-7)	MPU Interrupt Request IRQA (IRQB)
0	0	↓ Active	Set high on ↓ of CA1 (CB1)	Disabled — $\overline{\text{IRQ}}$ remains high
0	1	↓ Active	Set high on ↓ of CA1 (CB1)	Goes low when the interrupt flag bit CRA-7 (CRB-7) goes high
1	0	↑ Active	Set high on ↑ of CA1 (CB1)	Disabled — $\overline{\text{IRQ}}$ remains high
1	1	↑ Active	Set high on ↑ of CA1 (CB1)	Goes low when the interrupt flag bit CRA-7 (CRB-7) goes high

- Notes: 1. ↑ indicates positive transition (low to high)
 2. ↓ indicates negative transition (high to low)
 3. The Interrupt flag bit CRA-7 is cleared by an MPU Read of the A Data Register, and CRB-7 is cleared by an MPU Read of the B Data Register.
 4. If CRA-0 (CRB-0) is low when an interrupt occurs (Interrupt disabled) and is later brought high, $\overline{\text{IRQA}}$ ($\overline{\text{IRQB}}$) occurs after CRA-0 (CRB-0) is written to a "one".



Control of CA1 and CB1 Interrupt Input Lines (CRA-0, CRB-0, CRA-1, and CRB-1) — The two lowest order bits of the control registers are used to control the interrupt input lines CA1 and CB1. Bits CRA-0 and CRB-0 are

used to enable the MPU interrupt signals $\overline{\text{IRQA}}$ and $\overline{\text{IRQB}}$, respectively. Bits CRA-1 and CRB-1 determine the active transition of the interrupt input signals CA1 and CB1 (Table 3).

TABLE 4 — CONTROL OF CA2 AND CB2 AS INTERRUPT INPUTS
CRA5 (CRB5) is low

CRA-5 (CRB-5)	CRA-4 (CRB-4)	CRA-3 (CRB-3)	Interrupt Input CA2 (CB2)	Interrupt Flag CRA-6 (CRB-6)	MPU Interrupt Request $\overline{\text{IRQA}}$ ($\overline{\text{IRQB}}$)
0	0	0	↓ Active	Set high on ↓ of CA2 (CB2)	Disabled — $\overline{\text{IRQ}}$ re- mains high
0	0	1	↓ Active	Set high on ↓ of CA2 (CB2)	Goes low when the interrupt flag bit CRA-6 (CRB-6) goes high
0	1	0	↑ Active	Set high on ↑ of CA2 (CB2)	Disabled — $\overline{\text{IRQ}}$ re- mains high
0	1	1	↑ Active	Set high on ↑ of CA2 (CB2)	Goes low when the interrupt flag bit CRA-6 (CRB-6) goes high

- Notes:
- ↑ indicates positive transition (low to high)
 - ↓ indicates negative transition (high to low)
 - The Interrupt flag bit CRA-6 is cleared by an MPU Read of the A Data Register and CRB-6 is cleared by an MPU Read of the B Data Register.
 - If CRA-3 (CRB-3) is low when an interrupt occurs (Interrupt disabled) and is later brought high, $\overline{\text{IRQA}}$ ($\overline{\text{IRQB}}$) occurs after CRA-3 (CRB-3) is written to a "one".

TABLE 5 — CONTROL OF CB2 AS AN OUTPUT
CRB-5 is high

CRB-5	CRB-4	CRB-3	CB2	
			Cleared	Set
1	0	0	Low on the positive transition of the first E pulse following an MPU Write "B" Data Register operation.	High when the interrupt flag bit CRB-7 is set by an active transition of the CB1 signal.
1	0	1	Low on the positive transition of the first E pulse after an MPU Write "B" Data Register operation.	High on the positive edge of the first "E" pulse following an "E" pulse which occurred while the part was deselected.
1	1	0	Low when CRB-3 goes low as a result of an MPU Write in Control Register "B".	Always low as long as CRB-3 is low. Will go high on an MPU Write in Control Register "B" that changes CRB-3 to "one".
1	1	1	Always high as long as CRB-3 is high. Will be cleared when an MPU Write Control Register "B" results in clearing CRB-3 to "zero".	High when CRB-3 goes high as a result of an MPU Write into Control Register "B".



Control of CA2 and CB2 Peripheral Control Lines (CRA-3, CRA-4, CRA-5, CRB-3, CRB-4, and CRB-5) — Bits 3, 4, and 5 of the two control registers are used to control the CA2 and CB2 Peripheral Control lines. These bits determine if the control lines will be an interrupt input or an output control signal. If bit CRA-5 (CRB-5)

is low, CA2 (CB2) is an interrupt input line similar to CA1 (CB1) (Table 4). When CRA-5 (CRB-5) is high, CA2 (CB2) becomes an output signal that may be used to control peripheral data transfers. When in the output mode, CA2 and CB2 have slightly different characteristics (Tables 5 and 6).

TABLE 6 — CONTROL OF CA-2 AS AN OUTPUT
CRA-5 is high

CRA-5	CRA-4	CRA-3	CA2	
			Cleared	Set
1	0	0	Low on negative transition of E after an MPU Read "A" Data operation.	High when the interrupt flag bit CRA-7 is set by an active transition of the CA1 signal.
1	0	1	Low on negative transition of E after an MPU Read "A" Data operation.	High on the negative edge of the first "E" pulse which occurs during a deselect.
1	1	0	Low when CRA-3 goes low as a result of an MPU Write to Control Register "A".	Always low as long as CRA-3 is low. Will go high on an MPU Write to Control Register "A" that changes CRA-3 to "one".
1	1	1	Always high as long as CRA-3 is high. Will be cleared on an MPU Write to Control Register "A" that clears CRA-3 to a "zero".	High when CRA-3 goes high as a result of an MPU Write to Control Register "A".

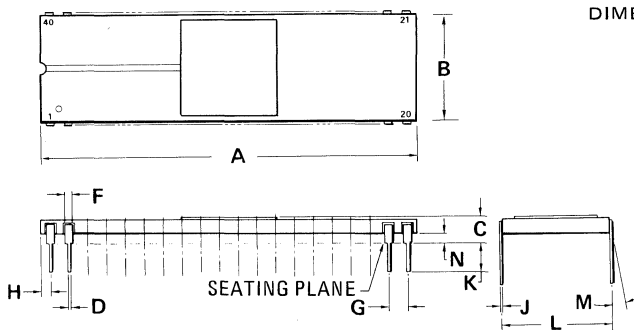
PIN ASSIGNMENT

1	O	CA1	40
2	V _{SS}	CA2	39
3	PA1	IRQA	38
4	PA2	IRQB	37
5	PA3	RS0	36
6	PA4	RS1	35
7	PA5	Reset	34
8	PA6	D0	33
9	PA7	D1	32
10	PB0	D2	31
11	PB1	D3	30
12	PB2	D4	29
13	PB3	D5	28
14	PB4	D6	27
15	PB5	D7	26
16	PB6	E	25
17	PB7	CS1	24
18	CB1	CS2	23
19	CB2	CS0	22
20	V _{CC}	R/W	21

PACKAGE DIMENSIONS

CASE 715-02
(CERAMIC)

SEE PAGE 165 FOR
PLASTIC PACKAGE
DIMENSIONS.



DIM	MILLIMETERS		INCHES	
	MIN	MAX	MIN	MAX
A	50.29	51.31	1.980	2.020
B	14.86	15.62	0.585	0.615
C	2.54	4.19	0.100	0.165
D	0.38	0.53	0.015	0.021
F	0.76	1.40	0.030	0.055
G	2.54 BSC		0.100 BSC	
H	0.76	1.78	0.030	0.070
J	0.20	0.33	0.008	0.013
K	2.54	4.19	0.100	0.165
L	14.60	15.37	0.575	0.605
M	—	10°	—	10°
N	0.51	1.52	0.020	0.060

NOTE:

1. LEADS, TRUE POSITIONED WITHIN 0.25 mm (0.010) DIA (AT SEATING PLANE), AT MAX. MAT'L CONDITION.

